

CyberSentinel AI Executive Threat Assessment Report

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1. Executive Summary

CyberSentinel AI conducted an AI-assisted defensive security assessment on 2026-08-20 17:49:57 UTC. A total of 4 vulnerability findings were identified across analyzed endpoints. This includes 1 Critical, 2 High, 1 Medium, and 0 Low severity items. Immediate remediation action is recommended for all Critical and High findings to reduce platform risk exposure.

2. Assessment Overview & Methodology

This automated security evaluation utilizes CyberSentinel AI's multi-layered Deep Learning architecture. Traffic flow vectors are analyzed across 32 numerical attributes using Artificial Neural Networks (ANN/MLP), 1D Convolutional Neural Networks (1D CNN), and LSTM temporal sequence models trained on defensive CIC-IDS security benchmarks. Predictions are augmented with a transparent risk scoring engine and feature importance explainability.

3. Severity Distribution Summary

Total Findings	Critical	High	Medium	Low
4	1	2	1	0

4. Detailed Technical Findings & Remediation

1. [CRITICAL] Volumetric HTTP Flood Attack Detected (DoS) (FIND-2026-001)

Category: Denial of Service (DoS) | **Confidence:** 99.2% | **Asset:** Gateway Load Balancer / API Gateway (10.0.4.15) | **Status:** New

Description: AI Detection Engine identified a sudden spike of 45,000 requests/sec with TCP SYN flag anomaly originating from external subnet 198.51.100.0/24.

Potential Impact: Exhaustion of connection table capacity leading to service outage for legitimate API clients.

Recommended Remediation: 1. Enable AWS Shield / Cloudflare DDoS Protection. 2. Configure IP rate-limit rule: max 200 req/min per IP. 3. Drop TCP SYN packets without valid cookie response.

2. [HIGH] SQL Injection Vector in User Authentication Parameter (FIND-2026-002)

Category: Web Attack - SQLi | **Confidence:** 94.5% | **Asset:** Auth Microservice /api/v1/auth/login | **Status:** Under Review

Description: Malicious SQL payload `` UNION SELECT username, password_hash FROM users--` detected in request body parameter during AI payload inspection.

Potential Impact: Unauthorized data extraction from backend PostgreSQL database tables and credential bypass.

Recommended Remediation: 1. Replace raw SQL query with SQLAlchemy parameterized ORM calls. 2. Apply input validation regex restricting special characters in username fields.

3. [HIGH] SSH Automated Brute Force Campaign (FIND-2026-003)

Category: Brute Force | **Confidence:** 96.1% | **Asset:** Bastion Host (192.168.1.10:22) | **Status:** Resolved

Description: Sustained sequence of 1,200 failed SSH authentication events detected over 300 seconds from remote host 203.0.113.45.

Potential Impact: Potential unauthorized shell access if weak SSH credentials exist on target bastion host.

Recommended Remediation: 1. Set 'PasswordAuthentication no' in /etc/ssh/sshd_config. 2. Restrict SSH access to authorized corporate VPN IP range.

4. [MEDIUM] Automated Port Scan Sequence Detected (FIND-2026-004)

Category: PortScan | **Confidence:** 89.7% | **Asset:** Internal Subnet Range (10.0.1.0/24) | **Status:** Resolved

Description: Sequential TCP SYN packets directed to 1,024 destination ports within 15 seconds indicating automated Nmap scan activity.

Potential Impact: Reconnaissance phase of multi-stage attack enabling discovery of open service ports.

Recommended Remediation: 1. Configure iptables to drop TCP port sweep attempts. 2. Enable automatic dynamic IP blocking.

5. Ethical Boundaries & Scope Statement

CyberSentinel AI operates strictly as an educational and defensive security threat analysis platform. It does not perform autonomous exploitation, credential theft, unauthorized network scanning, or payload execution. All intelligence generated herein is intended solely for security posture hardening and risk mitigation.